EXPERIMENT NO: -04

AIM OF THE EXPERIMENT; - TO STUDY THE SPEED CONTROL OF A D.C SHUNT MOTOR BY FLUX CONTROL AND ARAMUTURE CONTROL METHOD.

THEORY; -

D.C motor is a device, which convert electrical energy to mechanical energy.

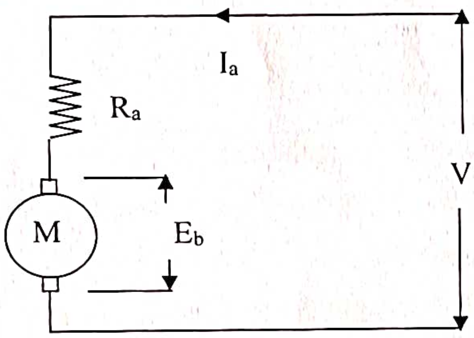
D.C shunt motor is a device whose field winding is in parallel with the armature winding.

If the DC motor is running at a speed N and the flux is ϕ, then the back emf developed in the machine is given by $E_b = \frac{\phi Z N}{60} \times \frac{P}{A}$

$= K\phi N$, Where $K = \frac{Z \times P}{60\ A}$ is a constant

The circuit equation of a DC shunt motor can be written as

$$V = E_b + I_a R_a$$



$\therefore\ E_b = K\phi N$

$\Rightarrow V - I_a R_a = K\phi N$

$N = (V - I_a R_a)/ K\phi$

By controlling the field flux ϕ($\phi \propto I_f$), the speed can be controlled. This is called flux **control method**

By adding external variable resistance in the armature circuit as per the expression

$N = (V - I_a (R_a + R_{EXT}))/ K\phi$

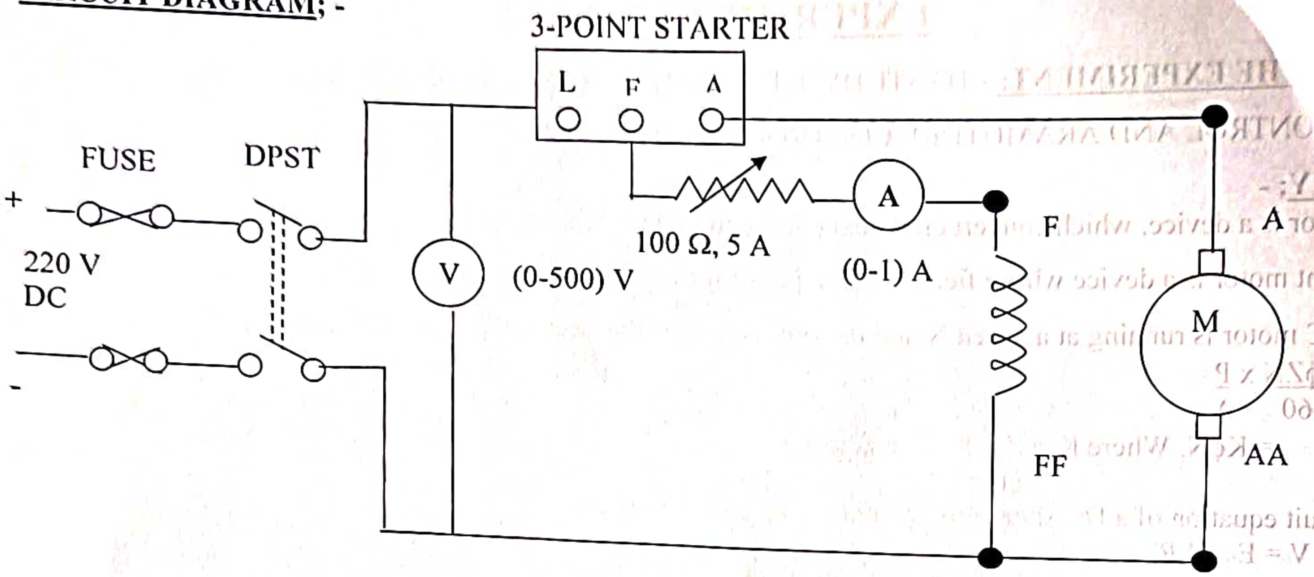
The speed could be controlled. This is called **Armature control method**

1. **FLUX CONTROL METHOD; -**

APPARATUS REQUIRED; -

SL. NO	APPARATUS	SPECIFICATION	QUANTITY
1	D.C SHUNT MOTOR	4 HP, 220 V, 16 A	01
2	TECHOMETER	(1000-4000) R.P.M	01
3	AMMETER	(0-1) A, PMMC TYPE	01
4	RHEOSTAT	(0-100)Ω, 1.5A	01
5	CONNECTING WIRES	3/22 SWG, INSULATED CU WIRES	AS REQUIRED

CIRCUIT DIAGRAM; -



PROCEDURE; -

1. MAKE THE CONNECTION AS PER THE CIRCUIT DIAGRAM.
2. KEEP THE RHEOSTAT IN ZERO POSITION AND SWITCH ON THE SUPPLY. INCREASE THE VOLTAGE FROM ZERO TO ITS RATED VALUE.
3. TAKE THE READING OF FIELD CURRENT AND SPEED OF THE MOTOR.
4. VARY THE VALUE OF THE RESISTANCE GRADUALLY AND TAKE THE CORRESPONDING READING OF FIELD CURRENT AND SPEED.
5. REPEAT STEP-4 AND TAKE 8 TO 10 READINGS.
6. PLOT THE GRAPH BETWEEN SPEED VS. FIELD CURRENT

TABULATION; -

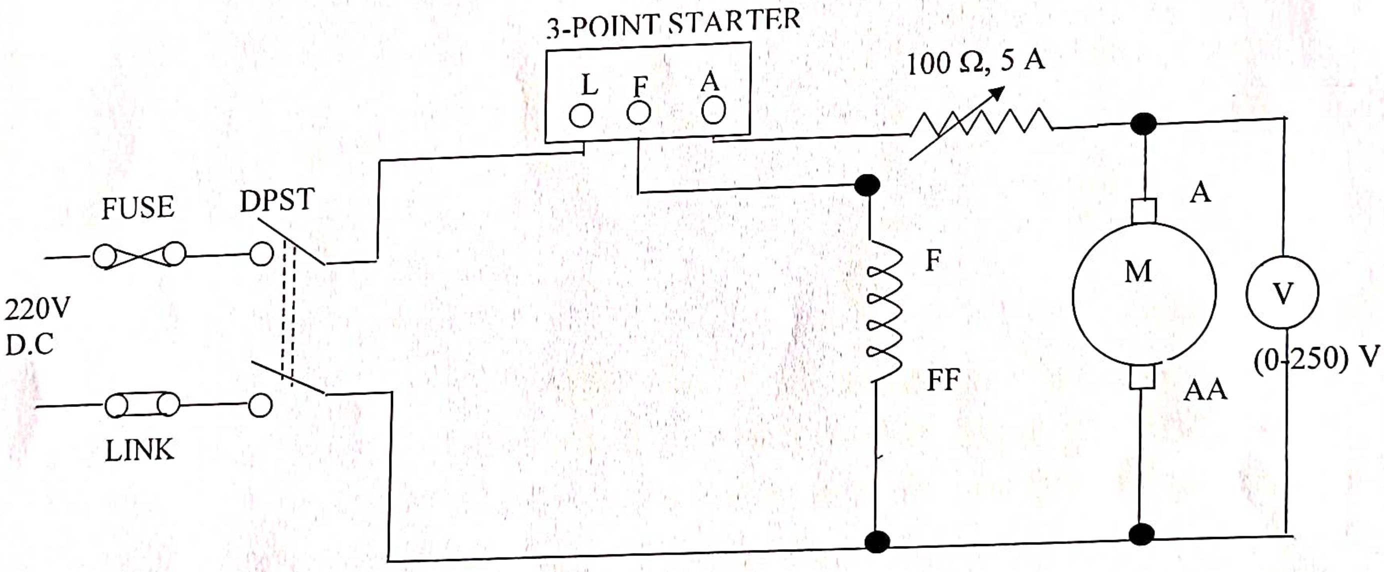
SL. NO.	FIELD CURRENT (I_F) IN AMP.	SPEED OF ARMATURE (N) IN R.P.M
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		

2. **ARMATURE CONTROL METHOD; -**

APPRATUS REQUIRED; -

SL. NO	APPARATUS	SPECIFICATION	QUANTITY
1	D.C SHUNT MOTOR	4 HP, 220 V, 16 A	01
2	TECHOMETER	(1000-4000) R.P.M	01
3	AMMETER	(0-1) A, PMMC TYPE	01
4	RHEOSTAT	(0-100)Ω, 1.5A	01
5	CONNECTING WIRES	3/22 SWG, INSULATED CU WIRES	AS REQUIRED

CIRCUIT DIAGRAM: -



PROCEDURE: -

1. MAKE THE CONNECTION AS PER THE CIRCUIT DIAGRAM.
2. KEEP THE POTENTIOMETER IN ZERO POSITION AND SWITCH ON THE SUPPLY. INCREASE THE VOLTAGE FROM ZERO TO ITS RATED VALUE.
3. TAKE THE READING OF FIELD CURRENT AND SPEED OF THE MOTOR.
4. VARY THE VALUE OF THE RESISTANCE GRADUALLY AND TAKE THE CORRESPONDING READING OF ARMATURE VOLTAGE AND SPEED.
5. REPEAT STEP-4 AND TAKE 8 TO 10 READINGS.
6. PLOT THE GRAPH BETWEEN SPEED VS. ARMATURE VOLTAGE.

TABULATION; -

SL. NO.	READING OF VOLT METER (V) IN VOLT.	SPEED OF ARMATURE (N) IN R.P.M
1.		
2.		
3.		
4.		
5.		
6		
7.		
8		

RESULT: -

Draw the two curves between

i. Speed vs. field current (N vs. I_f)

ii. Speed vs. armature voltage (N vs. V)

CONCLUSION: -

PRECAUTION; -

1. ALL THE CONNECTION SHOULD BE RIGHT AND TIGHT.
2. THE VOLTMETER AND THE AMMETER SHOULD BE CAREFULLY CHOSEN, SO THAT THEIR RANGES ARE MORE THAN THE MAXIMUM VALUES TO BE MEASURED.
3. WHILE MOVING THE STARTER HANDLE TO ITS FINAL POSITION, MOVE SLOWLY.
4. SUPPLY SHOULD BE SWITCH ON AFTER ENSURING CORRECTNESS OF CONNECTION.
5. AFTER SWITCHING OFF THE D.C SUPPLY, CHEACK THAT STARTER HANDLE COMES BACK TO ITS OFF POSITION.
6. ANY LIVE TERMINAL SHOULD NOT BE TOUCHED, WHILE SUPPLY IS ON.